

MongoDB CRUD Operations using Command Prompt

1. Database & Collection Creation

```
Windows PowerShell
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PS C:\Users\AARTHI> mongosh
Current Mongosh Log ID: 69f753e48e2a21c051abc113
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelection
TimeoutMS=2000&appName=mongosh+2.8.3
Using MongoDB:      8.2.7
Using Mongosh:       2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

-----
The server generated these startup warnings when booting
2026-05-03T12:51:04.716+05:30: Access control is not enabled for the database. Read and
write access to data and configuration is unrestricted
-----
```

2. Insert Operations

```
test> use studentDB
switched to db studentDB
studentDB> db.createCollection("students")
{ ok: 1 }
studentDB> db.students.insertOne({
|   name: "Aarthi",
|   age: 21,
|   course: "MERN Stack",
|   status: "ongoing"
| })
{
  acknowledged: true,
  insertedId: ObjectId('69f754958e2a21c051abc114')
}
studentDB> db.books.insertMany([
| {
|   title: "MongoDB Basics",
|   author: "John",
|   category: "Database",
|   price: 500,
|   available: true
| },
| {
|   title: "Node.js Guide",
|   author: "David",
|   category: "Programming",
|   price: 700,
|   available: true
| },
| {
|   title: "React Handbook",
|   author: "Steve",
|   category: "Frontend",
|   price: 650,
|   available: false
| }
| ])
{
}
```

3. Read Operations

```
studentDB> db.students.find(
|   {},
|   { name: 1, course: 1, _id: 0 }
| )
[
|   { name: 'Aarthi', course: 'MERN Stack' },
|   { name: 'Rahul', course: 'Python' },
|   { name: 'Priya', course: 'MERN Stack' },
|   { name: 'Kavin', course: 'Java' }
| ]
studentDB> db.students.updateOne(
|   { name: "Aarthi" },
|   { $set: { status: "completed" } }
| )
{
|   acknowledged: true,
|   insertedId: null,
|   matchedCount: 1,
|   modifiedCount: 0,
|   upsertedCount: 0
| }
studentDB> db.students.updateMany(
|   { status: "ongoing" },
|   { $set: { status: "completed" } }
| )
{
|   acknowledged: true,
|   insertedId: null,
|   matchedCount: 1,
|   modifiedCount: 1,
|   upsertedCount: 0
| }
studentDB> db.students.deleteOne({ name: "Kavin" })
{ acknowledged: true, deletedCount: 1 }
studentDB> db.students.deleteMany({ status: "pending" })
{ acknowledged: true, deletedCount: 0 }
studentDB> db.students.find({ age: { $gt: 21 } })
[
```

```

studentDB> db.students.find({ course: "MERN Stack" })
[
  {
    _id: ObjectId('69f754958e2a21c051abc114'),
    name: 'Aarthi',
    age: 21,
    course: 'MERN Stack',
    status: 'completed'
  },
  {
    _id: ObjectId('69f758c48e2a21c051abc119'),
    name: 'Priya',
    age: 20,
    course: 'MERN Stack',
    status: 'ongoing'
  }
]
studentDB> db.students.find(
  |   {},
  |   { name: 1, course: 1, _id: 0 }
  | )
[
  { name: 'Aarthi', course: 'MERN Stack' },
  { name: 'Rahul', course: 'Python' },
  { name: 'Priya', course: 'MERN Stack' },
  { name: 'Kavin', course: 'Java' }
]
studentDB> db.students.updateOne(
  |   { name: "Aarthi" },
  |   { $set: { status: "completed" } }
  | )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}

```

```
,
studentDB> db.students.find()
[
  {
    _id: ObjectId('69f754958e2a21c051abc114'),
    name: 'Aarthi',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  }
]
studentDB>

studentDB> db.students.find().pretty()
[
  {
    _id: ObjectId('69f754958e2a21c051abc114'),
    name: 'Aarthi',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  }
]
studentDB> db.students.find({ course: "MERN Stack" })
[
  {
    _id: ObjectId('69f754958e2a21c051abc114'),
    name: 'Aarthi',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  }
]
```

4. Update Operation

```
studentDB> db.students.updateMany(
|   { status: "ongoing" },
|   { $set: { status: "completed" } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentDB> db.students.deleteOne({ name: "Kavin" })
{ acknowledged: true, deletedCount: 1 }
studentDB> db.students.deleteMany({ status: "pending" })
{ acknowledged: true, deletedCount: 0 }
studentDB> db.students.find({ age: { $gt: 21 } })
[
  {
    _id: ObjectId('69f758c48e2a21c051abc118'),
    name: 'Rahul',
    age: 22,
    course: 'Python',
    status: 'completed'
  }
]
studentDB> db.students.find({ course: "MERN Stack" })
[
  {
    _id: ObjectId('69f754958e2a21c051abc114'),
    name: 'Aarthi',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  }
]
studentDB> db.students.updateOne(
|   { name: "Aarthi" },
|   { $set: { status: "completed" } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

5. Delete Operations

```
studentDB> db.students.deleteOne({ name: "Rahul" })
{ acknowledged: true, deletedCount: 0 }
studentDB> db.students.find()
[
  {
    _id: ObjectId('69f754958e2a21c051abc114'),
    name: 'Aarthi',
    age: 21,
    course: 'MERN Stack',
    status: 'completed'
  }
]
studentDB> db.students.deleteOne({ name:"John" })
{ acknowledged: true, deletedCount: 0 }
studentDB> db.students.find({ age: { $gt: 20 } })
[
  {
    _id: ObjectId('69f754958e2a21c051abc114'),
    name: 'Aarthi',
    age: 21,
    course: 'MERN Stack',
    status: 'completed'
  }
]
studentDB> db.students.find({ age: { $lt: 22 } })
[
  {
    _id: ObjectId('69f754958e2a21c051abc114'),
    name: 'Aarthi',
    age: 21,
    course: 'MERN Stack',
    status: 'completed'
  }
]
```

6. Query Operators

```
studentDB> db.students.find({ age: { $lt: 22 } })
[
  {
    _id: ObjectId('69f754958e2a21c051abc114'),
    name: 'Aarthi',
    age: 21,
    course: 'MERN Stack',
    status: 'completed'
  },
  {
    _id: ObjectId('69f758c48e2a21c051abc119'),
    name: 'Priya',
    age: 20,
    course: 'MERN Stack',
    status: 'completed'
  }
]
```

```
studentDB> db.students.find(
|   course: { $in: ["MERN Stack", "Python"] }
| })
[
  {
    _id: ObjectId('69f754958e2a21c051abc114'),
    name: 'Aarthi',
    age: 21,
    course: 'MERN Stack',
    status: 'completed'
  },
  {
    _id: ObjectId('69f758c48e2a21c051abc118'),
    name: 'Rahul',
    age: 22,
    course: 'Python',
    status: 'completed'
  },
  {
    _id: ObjectId('69f758c48e2a21c051abc119'),
    name: 'Priya',
    age: 20,
    course: 'MERN Stack',
    status: 'completed'
  }
]
```



```

studentDB> db.students.find({
|     $and: [
|         { age: { $gt: 20 } },
|         { status: "completed" }
|     ]
| })
[
| {
|   _id: ObjectId('69f754958e2a21c051abc114'),
|   name: 'Aarthi',
|   age: 21,
|   course: 'MERN Stack',
|   status: 'completed'
| },
| {
|   _id: ObjectId('69f758c48e2a21c051abc118'),
|   name: 'Rahul',
|   age: 22,
|   course: 'Python',
|   status: 'completed'
| }
| ]
| }
| ]

```

```

studentDB> db.students.find({
|     $or: [
|         { course: "MERN Stack" },
|         { course: "Java" }
|     ]
| })
[
| {
|   _id: ObjectId('69f754958e2a21c051abc114'),
|   name: 'Aarthi',
|   age: 21,
|   course: 'MERN Stack',
|   status: 'completed'
| },
| {
|   _id: ObjectId('69f758c48e2a21c051abc119'),
|   name: 'Priya',
|   age: 20,
|   course: 'MERN Stack',
|   status: 'completed'
| }
| ]
| ]

```

```
studentDB> db.students.find({
|   status: { $exists: true }
| })
[
|   {
|     _id: ObjectId('69f754958e2a21c051abc114'),
|     name: 'Aarthi',
|     age: 21,
|     course: 'MERN Stack',
|     status: 'completed'
|   },
|   {
|     _id: ObjectId('69f758c48e2a21c051abc118'),
|     name: 'Rahul',
|     age: 22,
|     course: 'Python',
|     status: 'completed'
|   },
|   {
|     _id: ObjectId('69f758c48e2a21c051abc119'),
|     name: 'Priya',
|     age: 20,
|     course: 'MERN Stack',
|     status: 'completed'
|   }
| ]
```

Conclusion

This exercise covers MongoDB basics like creating a database and collection, inserting data, reading data, updating records, and deleting data. It also includes simple query operators like \$gt, \$lt, \$in, \$and, \$or, and \$exists. Overall, it helps to understand how to manage data in MongoDB using mongosh.